Rush Hour: Predicting Bike-Sharing Demand

Chained pipeline report · Team IDs 16006349, 16009603 · seed = 36662 (80/20 split: 8,708 train / 2,178 validation)

Data exploration (before Phase 1)

We explored train.csv before modeling, and every Phase 1 decision below rests on these findings. **Time structure:** average hourly rentals grew from 144.2 (2011) to 238.6 (2012), with a seasonal wave. Working days show two commuter peaks (about 480 bikes at 8am and 529 at 5pm, dipping to 134 at 10am), while weekends show one midday hump (peak about 388), so the effect of the hour depends on the day type. **Target:** cnt is right-skewed (skewness 1.24, maximum 977, no zeros). **Redundancy hypotheses:** temp and atemp correlate at 0.985; every season contains exactly three months; workingday is reproduced exactly by weekday and holiday in 100% of rows; instant correlates 1.0 with the date. **Quirks:** windspeed is exactly 0 in 1,313 rows (likely missing values), hum is 0 in 22 rows, weathersit = 4 occurs once, and 42 days have missing hours. **Test structure:** train.csv covers days 1–19 of each month and test.csv is exactly the 20th of all 24 months, so the hidden test consists of unseen whole days spread across both years. This shaped our Phase 3 analysis.

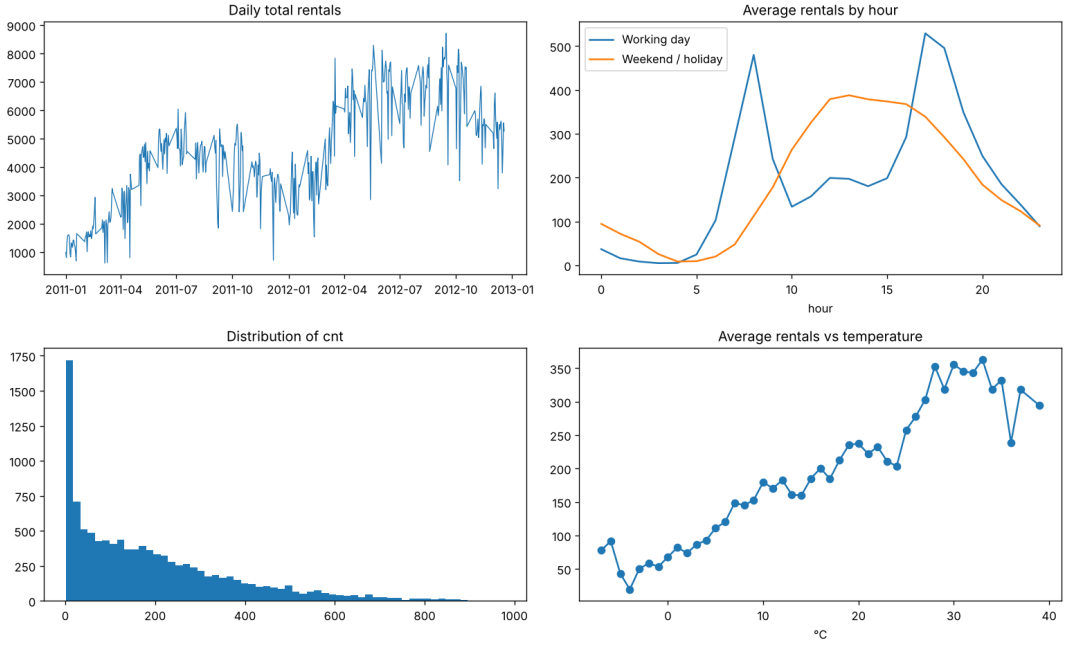


Figure 1. Daily rentals, average rentals by hour and day type, distribution of cnt, and rentals vs temperature.

Phase 1 – Gradient descent from scratch

Features (75). hr, mnth, weekday, season and weathersit are categories, not quantities, so they are one-hot encoded with one baseline category dropped. A numeric hr would force a straight-line hour effect that cannot draw two peaks. We added an **hr × workingday** interaction (23 columns) so working days and weekends get different daily shapes. weathersit = 4 (one row) was merged into 3. instant and dteday were not used: dteday's useful content (year, month) is already in yr and mnth. Redundant columns (atemp, season, weekday) were deliberately kept so Phase 4 could test them with evidence. **Scaling:** every feature is standardized with the training mean and standard deviation only, then applied unchanged to validation and test. **Target:** the model predicts log(1 + cnt) and scores are computed after converting back to bike counts. The same model trained on raw cnt gave validation R^2 0.826, RMSE 74.6 and 258 negative predictions; the log model gave R^2 0.871 and no negative predictions.

Learning rate and stopping. We stop when the relative loss improvement drops below 10^{-7} (cap 10,000 iterations). A sweep showed lr = 0.001 and 0.01 still improving at the cap, lr = 0.1 converging on its own in 2,071 iterations (training MSE 0.1220), and lr = 0.3 diverging after 68 iterations. This matches theory: the largest stable rate, 2 divided by the largest eigenvalue of the loss curvature matrix, is 0.273 for our data. We chose **lr = 0.1**. As a correctness check, gradient descent matches the exact least-squares solution (identical validation R^2 0.8708; maximum prediction difference 0.016 on the log scale). **Result:** train R^2 0.889 / RMSE 60.7, validation R^2 **0.871** / RMSE **64.4**. **Passed to Phase 2:** the 76-element weight vector (bias + 75), lr = 0.1, 2,071 iterations.

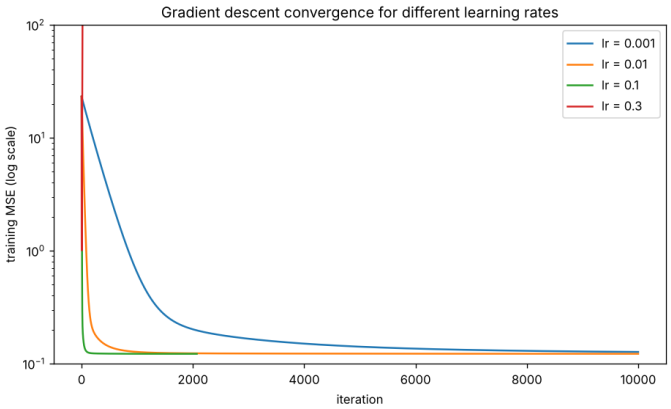


Figure 2. Training loss per iteration for four learning rates.

Bonus – Asymmetric loss (under-prediction costs 3×)

With error $e_i = x_i \cdot w - y_i$ and weight $c_i = 3$ if $e_i < 0$, otherwise 1, the loss is $L(w) = (1/n)\ \Sigma\ c_i\ e_i^2$. Because c_i is constant on each side of zero, the gradient is $\nabla L(w) = (2/n)\ X^T(c \odot e)$. Both sides give derivative 0 at $e_i = 0$, so the loss is smooth and convex and plain gradient descent applies. The weight can triple the gradient, lowering the stability limit to about 0.273 / 3 = 0.091, so we used lr = 0.05 (converged in 2,213 iterations) with the same features as Phase 1. **Shift:** mean prediction rose from 184.9 to 204.3 bikes (actual 188.8), and the share of under-predicted hours fell from 55.5% to 38.4%. R^2 fell to 0.842 and RMSE rose to 71.2, as expected, since squared error is no longer the goal, but the operator's asymmetric cost on validation fell by **11%** (7,886 → 6,990 bikes²). Because the model works on the log scale, the shift is proportional: almost none at night, but 8am working-day predictions rose from 468 to 524 (actual 477) and 5pm from 557 to 595 (actual 545). The model now aims at the 0.75 expectile instead of the mean.

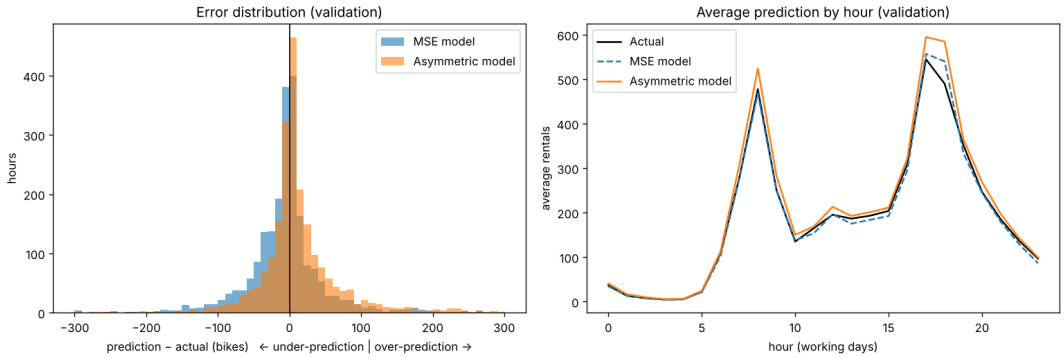


Figure 3. Validation errors and average working-day predictions: MSE model vs asymmetric model.

Phase 2 – Polynomial regression

Which features. We expanded only the continuous weather columns temp, hum and windspeed, including their cross terms (e.g. temp·hum), because curves are physically plausible there: Figure 1 shows demand rising with temperature and flattening when it is very hot. Squaring 0/1 columns creates exact duplicates ($1^2 = 1$), and atemp would duplicate temp's curves. **Initialization:** the original 75 columns keep Phase 1's scaling, so Phase 1's 76 weights are copied into their positions and the new polynomial weights start at 0. The expanded model therefore starts exactly at Phase 1's loss (0.1220); degree 1 stops after 2 iterations, confirming the chain. For each degree the learning rate is min(0.1, 0.9 × stability limit).

Degree	Features	lr	Iterations	Train R^2	Val R^2	Val RMSE
1	75	0.100	2	0.8885	0.8708	64.36
2	81	0.100	3,460	0.9082	0.8998	56.68
3 (chosen)	91	0.100	12,751	0.9116	0.9047	55.26
4	106	0.060	20,000 (cap)	0.9123	0.9053	55.10
5	127	0.037	20,000 (cap)	0.9128	0.9059	54.92

Why degree 3. Degree 1 → 2 adds 0.029 validation R^2, and 2 → 3 adds 0.005 while still converging on its own at lr = 0.1. Degrees 4–5 add only about 0.001 in total, need smaller learning rates and hit the iteration cap, because x, x^2, x^3 and x^4 are strongly correlated and the loss becomes a long, narrow valley. The train–validation gap stays near 0.007 at every degree. **Passed to Phase 3:** degree 3 on temp, hum and windspeed (91 features).

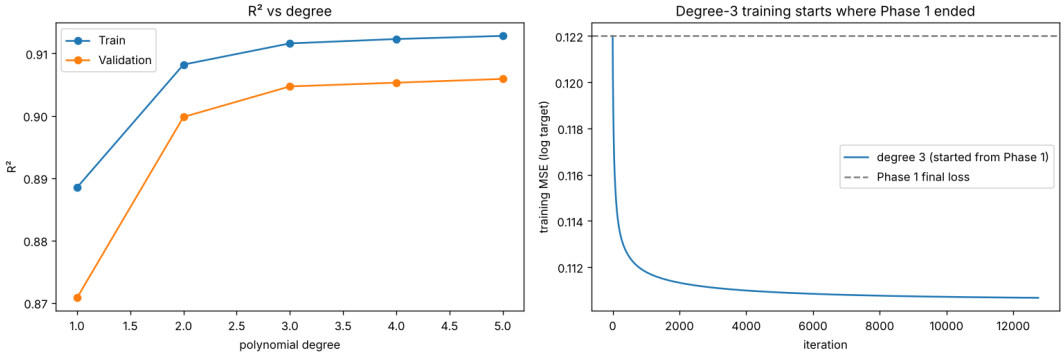


Figure 4. R^2 vs degree, and degree-3 training starting from Phase 1's final loss.

Phase 3 – Bias-variance tradeoff and the time question

What we varied. Complexity (degrees 1–5, from Phase 2) and training-set size (a learning curve at degree 3 using 10%–100% of the training rows). With 870 rows the train–validation gap is 0.043 (train 0.927, validation 0.885); from about 3,500 rows on it stays at 0.006–0.009 and both curves are flat near 0.905. More data does not help and the gap does not grow with degree, so the model is **not overfitting**; the remaining error is **bias** from the model's linear-in-log form.

Chronological split. We trained on all dates before 1 August 2012 (8,607 rows) and validated on August–December 2012 (2,279 rows, 21%, close to the seeded 20%), refitting the scaler on the earlier period only. At degree 3, validation R^2 was almost unchanged (0.904 vs 0.905) but RMSE rose from 55.3 to **67.6** bikes (+22%). R^2 is relative to the target's variance, and late 2012 is the busiest, most variable period, so R^2 hid the larger absolute error. Monthly averages show a systematic **over-prediction** in every month (e.g. August: 285.6 actual vs 307.2 predicted; mean error −15.6 bikes/hour): the model extrapolates the 2012 uplift learned from January–July, while late-2012 growth was weaker. The random split hides this because hours from those months are in training.

Which estimate to trust. The random split is optimistic because hours of the same day sit in both training and validation. Since the hidden test is the 20th of each month, we also trained on days 1–18 and validated on day 19 (deterministic, no seed): R^2 0.891, RMSE 59.5. We trust this for the hidden test, and the chronological estimate (RMSE ≈ 68, with over-prediction when growth changes) for true future deployment. **Diagnosis passed to Phase 4:** reasonably fit with slight bias, no overfitting; the chronological split reveals trend bias, not variance. Degree 3 is confirmed (3 → 5 adds 0.001 on the chronological split too), so we expect regularization to help selection more than accuracy.

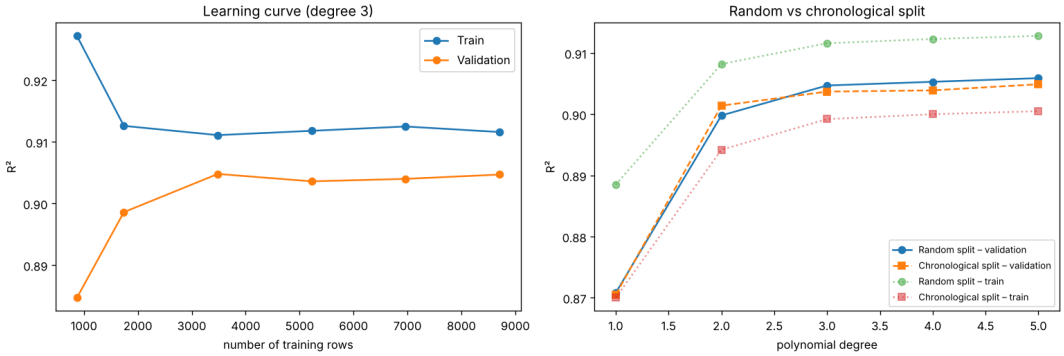


Figure 5. Learning curve at degree 3, and random vs chronological split across degrees.

Phase 4 – Regularization (Ridge, Lasso, Elastic Net)

All three methods are implemented in our own gradient descent on the 91 degree-3 features. Ridge adds λw to the gradient; Lasso uses **proximal gradient descent** (soft-thresholding each weight by $lr \cdot \lambda$ after every step) so weights reach exactly zero; Elastic Net combines both via l1_ratio. The bias is never penalized. **Search:** λ on a log grid from 10^{-5} to 10^{-1} (8 values), l1_ratio $\in \{0.2, 0.5, 0.8\}$ for Elastic Net, warm-started from Phase 2's weights. **Fair comparison:** same features, split and learning rate; each method judged at its own best λ on validation, ties broken toward the larger λ (simpler model). Results: Lasso $\lambda = 0.001$ (val R^2 0.9051, 82 of 91 weights non-zero), Elastic Net $\lambda = 0.0003$, l1_ratio 0.5 (0.9051, 91 non-zero), Ridge $\lambda = 0.001$ (0.9050, 91). As Phase 3 predicted, accuracy barely moved from Phase 2 (0.9047); performance falls for all methods beyond $\lambda \approx 0.003$ as the penalty causes underfitting.

Column verdicts. Lasso alone cannot identify redundant pairs (it may split weight between them), so we also ran drop-column tests at the chosen Lasso: remove every feature built from a column, retrain, and measure the change

in validation R^2. Redundancy is shown when dropping either column alone costs little but dropping both costs a lot.

Column	Verdict	Evidence from our models (change in validation R^2 when dropped)
hr	Useful	−0.680 (most important column)
workingday	Useful	−0.163 (including the hr × workingday interaction)
yr	Useful	−0.096 (2011 → 2012 growth)
temp	Useful	−0.024
hum	Useful	−0.009
weathersit	Useful	−0.004; both categories kept by Lasso at every λ
windspeed	Useful, weak	−0.002; Lasso zeroes its linear term at λ = 0.003; 1,313 zero readings
mnth	Useful	+0.0006 alone, but −0.020 together with season
atemp	Redundant (temp)	−0.0001; r = 0.985 with temp; temp + atemp together −0.025
season	Redundant (mnth)	0.0000; each season = exactly 3 months; season + mnth together −0.020
holiday	Redundant (workingday)	0.0000; first column Lasso removes completely (λ = 0.003)
weekday	Redundant (workingday)	+0.007 when dropped: helps log-MSE (+0.0035 worse without it) but hurts count-scale R^2
instant	Redundant (time)	adding it changes R^2 by −0.0001; correlation 1.0 with the date
dteday	Redundant as raw column	not usable as text; its year and month are already in yr and mnth

weekday is the subtle case: on the log scale it helps slightly, but small day-of-week log offsets are multiplied by the hour's demand when converted back to bikes, producing large absolute errors exactly at rush hours, which dominate count-scale R^2. Its useful content (working day vs weekend) is already in workingday. **Final model:** after removing atemp, season, holiday and weekday, all methods improved (Lasso λ = 0.001: train R^2 0.916, validation R^2 **0.912**, RMSE **53.2**, 75 of 80 weights non-zero; Elastic Net 0.912 with 80; Ridge 0.911). We recommend **Lasso λ = 0.001** because it ties on accuracy with the fewest features. It produced sample_submission.csv using exactly the same fitted transformations on test.csv; we did not refit on train + validation, to keep the reported validation numbers representative of the submitted model. **Passed to Phase 5:** 75 surviving features from 8 columns (hr, workingday, yr, mnth, weathersit, temp, hum, windspeed).

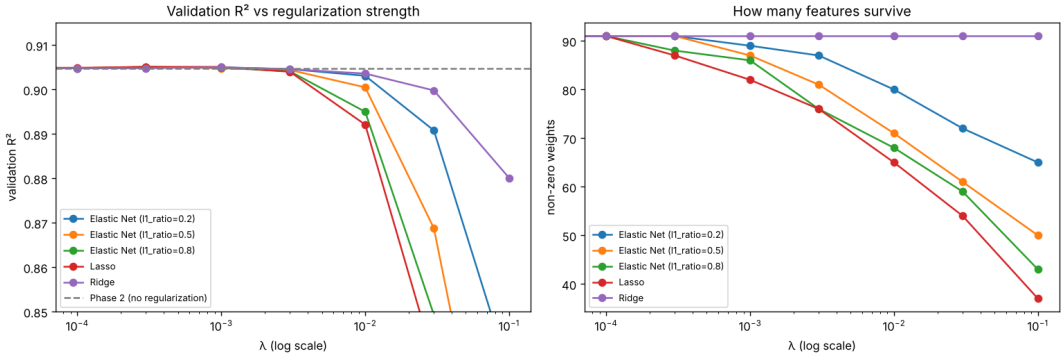


Figure 6. Validation R^2 and number of non-zero weights against λ for every method.

Phase 5 – Logistic regression

The flaw of a global threshold. Labelling an hour "high" when cnt exceeds the training median (145) marks 98% of working-day 8am and 5pm hours as high and 0% of 3am hours, and 59% of 2012 hours vs 40% of 2011 hours. A classifier using only the hour reaches validation AUC 0.906 on this label, so it encodes the clock, which the operator already knows. **Our definition:** an hour is high-demand if its cnt exceeds the training-set **75th percentile of its own slot** (same year × hour × working-day type), i.e. busier than three out of four comparable hours. Thresholds come from training data only. Fixing only hour × workingday exposed a second bias (year alone: AUC 0.759, from growth); adding the year removed both (hour-only AUC 0.523, year-only 0.491, both near chance). Positive share: 24.5% train, 26.1% validation.

Model. Logistic regression implemented with our own gradient descent on cross-entropy (gradient $X^T(p - y)/n$), using only Phase 4's 75 surviving features; lr = 1.0 converged in 2,086 iterations (loss 0.398), and sklearn, used only as a check, gives validation AUC 0.852 vs our 0.851. **Cut-off:** with 25% positives, 0.5 is too strict, so we chose the probability cut-off maximizing F1 on the **training** data (0.34) and applied it unchanged to validation. **Metrics:** accuracy is misleading (a model that never predicts high scores 0.739), so we report ROC-AUC and F1. Validation: ROC-AUC **0.851** (train 0.849, so no overfitting), F1 **0.638** (0.537 at a 0.5 cut-off), accuracy 0.776. Confusion

matrix: 430 of 569 high hours caught (76% recall), precision 55%, 349 false alarms, 139 misses. Missing a busy hour costs the operator more than a false alarm, so this trade-off suits the task. The strongest drivers are temperature terms and the July–October months, i.e. weather and season, not the clock.

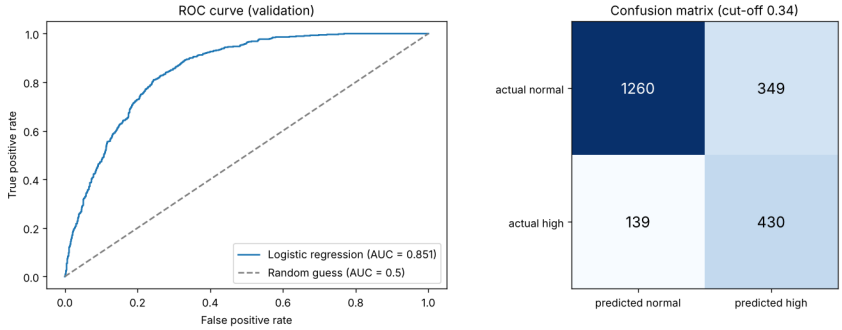


Figure 7. Validation ROC curve and confusion matrix at cut-off 0.34.

Pipeline retrospective

Phase	Consumed	Produced	Key numbers (validation)
1 Gradient descent	Seeded split (36662)	76 weights, lr 0.1, 2,071 iterations	R^2 0.871, RMSE 64.4; lr 0.3 diverges (limit 0.273)
Bonus	Phase 1 features	Asymmetric model, lr 0.05	Cost −11%; under-predicted 55.5% → 38.4%
2 Polynomial	Phase 1 weights as init	Degree 3 on temp, hum, windspeed (91)	R^2 0.905, RMSE 55.3; loss 0.1220 → 0.1107
3 Bias–variance	Degree 3	Slight bias, no overfitting; degree 3	Gap 0.007; chrono RMSE 67.6; day-19 R^2 0.891
4 Regularization	Diagnosis, degree 3	λ per method; 4 columns dropped; Lasso	R^2 0.912, RMSE 53.2; 75 features, 8 columns
5 Logistic	75 surviving features	Classifier, cut-off 0.34	ROC-AUC 0.851, F1 0.638, accuracy 0.776

Each phase's output directly shaped the next: Phase 0's two-peak finding drove the hr × workingday features; Phase 1's weights initialized Phase 2; Phase 3's "no overfitting" diagnosis correctly predicted that regularization would help feature selection more than accuracy; Phase 4's column verdicts removed four columns, which raised validation R^2 from 0.905 to 0.912 and defined Phase 5's inputs. The main limitation is trend extrapolation: for genuinely future periods the model over-predicts when growth slows, so a next step would be a model that tracks recent levels, or richer weather × hour interactions to reduce the remaining bias.